CITY COLLEGES

APPLIED MACHINE LEARNING

Final Project:

Applied Machine Learning for a FinTech Startup

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Assignment Tasks

1. Project Overview

Your task is to build a comprehensive **business case** for a new **data-driven personal finance or retail investment platform** (e.g., a budgeting app, robo-advisor, or trading assistant) and to design and implement a machine learning solution that support this business case. The goal is to simulate a realistic situation where a fintech start-up or financial institution wants to use machine learning to improve decision-making, reduce risk, and create value for users in financial markets or consumer finance. You will choose an appropriate public dataset (e.g., Kaggle, UCI, Yahoo Finance) and build:

- **One primary ML model** (classification or regression).
- **One comparison model** (baseline or alternative).
- **Explainability analysis using SHAP.**

Your goal is to demonstrate that you can take a financial dataset, clean it, engineer meaningful features, build an interpretable model, and evaluate it in a way that supports a realistic business scenario.

2. Problem Scenario

You will define a realistic problem in the personal finance / market domain and motivate why machine learning is suitable. Possible examples include:

- **Retail investment support:** Helping novice investors choose suitable stocks, ETFs, or portfolios based on their risk profile.
- **Market risk and volatility awareness:** Alerting users when their investments are exposed to abnormal risk or drawdowns.
- **Credit risk and responsible lending:** Assessing the risk of loan default for retail customers (e.g., credit scoring).
- **Fraud or anomaly detection:** Detecting suspicious card transactions or trading behaviour.
- **Savings and budgeting optimisation:** Helping users allocate monthly income across savings, debt repayment, and investments.
- **Client segmentation:** Grouping users into segments based on financial behaviour (e.g., risk-averse vs risk-seeking).

You may select another finance-related problem if it fits the assignment.You must clearly explain:

- The business problem,
- Why machine learning is suitable,
- Who benefits (e.g., bank, investor, customer).

3. Required ML Components

Identify a suitable public dataset (or combination of datasets) from a reliable source (e.g., Kaggle, UCI, Yahoo Finance API). Your dataset should be clearly linked to your problem and models. For example:

- Perform appropriate cleaning.
- Perform Exploratory Data Analysis (EDA).

a) One primary model

e.g., Random Forest, Gradient Boosting, Logistic Regres- sion, LSTM if time-series.

b) One comparison model.

You must justify:

- Why your chosen model fits the problem.
- What metrics are relevant.

c) Explainability (XAI)

Use SHAP to show to explain model decisions. Discuss how explainability improves:

- Trust,
- Model transparency,
- Business decision-making.

d) Evaluation

You must evaluate using any appropriate metrics, for example:

- Accuracy, precision, recall, F1, ROC-AUC (classification), RMSE, MAE, MAPE (regression).
- Graphs (e.g., confusion matrix, SHAP plot, feature importance).

1. Problem Description and Business Relevance

1.1 Business Context

Consumers face difficulty in managing their own financials especially in attempting to distribute their monthly income between spending, debt, and savings. Poor budgeting choices are the most significant contributors where overextending of credit, late payments and added financial pressure could ultimately result in credit default.

And for budgeting and financial wellness platforms like these fintech startups, helping users get better at where they direct their income is a civic responsibility as much as a business opportunity. In this project, we suggest a data-driven personal finance optimisation system to help users in tight financial condition by proactively identifying such users and recommend them better allocation of income for the purpose of repaying debt and saving.

By recontextualising credit default as a manifestation of poor budgeting and saving habits, this project aims to move beyond conventional scoring tools in borrowing towards proactive financial health management and early user intervention.

1.2 Business Problem

The fundamental business problem that the project aims to solve is detecting if a user carries high financial risk attributes;failing financial management and managing debt poorly in advance. That is, the platform is above all interested in answering the following question:

Can historical credit behavior and payment frequency be used to predict if an individual is at high risk of defaulting on credit obligations, requiring the intervention in budgeting and saving?

For instance, in their study, credit card default is considered to be an indicator of financial stress -there must be disporortionate allocation of income between consumption and (as teh creditors perceive) repayment of debt and savings.

1.3 Why The Machine Learning Approach?

The complex nature of financial behavior makes it a particularly suitable candidate for machine learning approaches. Both spending behaviour, credit usage and repayment history in conjunction with demographic data interact in complex non linear ways that are hard to grasp using rule based approaches.

In contrast to static thresholds or manual heuristics, supervised machine learning techniques have the ability to take cues from large historical datasets in order to detect early warning signals of financial distress. Moreover, automated and scalable decision making is required since our fintech platforms serve a broad spectrum of end-users. Machine learning can enable the platform to provide actionable, personalized budgeting and savings recommendations by predicting the chances of default for each user.

1.4 Stakeholders and Value Creation

This machine-based approach has already generated value to a number of parties:

- **Consumers:**
  Receive personalized guidance into their financial wellness, allowing them to make modifications to spending habits allocate funds toward debt and establish savings before financial struggles compound.
- **FinTech Startup:**
  Provides proactive and explainable financial guidance rather than reactive risk analysis to drive Improve user engagement, trust and retention.
- **Financial Institutions:**
  And benefit indirectly from reductions in default rates and improved responsible lending outcomes, generated by earlier intervention with the user.

2. Dataset Description and Cleaning

2.1 Dataset Overview

The dataset used in this project is the *Default of Credit Card Clients* dataset from the UCI Machine Learning Repository (Lichman, 2016). It has data of 30,000 credit card clients in Taiwan and it comprises a total of 25 variables that include demographic information, credit limits, historical bill amounts and repayments frequency, among others.

A first simple examination showed the dataset is free of missing values. Due to its widespread use in academic research in the evaluation of credit risk models, the dataset is per ceived as one that is trustworthy and of good quality for the study of default risk and financial behaviour.

2.2 Variables and Feature Groups

2.2.1 Identifier

- **ID**
  Unique identifier assigned to each client.
  This variable exists only for the purpose of record identification, and is excluded before training the model as it does not contribute to prediction.

2.2.2 Credit Limit

- **LIMIT_BAL**
  Total credit limits in NTD available for the cardholder, including primary and additional cardholders.
  This is a factor that represents access to credit and financial flexibility, factors that may affect spending plans and budgeting decisions.

2.2.3 Target Variable

- **default.payment.next.month**
  Indicates whether the client defaulted on their credit card payment in the next month:
  1 = Default
  0 = No default

This binary attribute would serve as a supervised classifier. Default risk in this project is taken as a proxy for financial distress and lack of direction of income towards repaying debt and savings. Accordingly, to achieve correct recognition of high-risk users, we prefer recall, F1-score and ROC-AUC than accuracy in terms of evaluation metrics.

2.2.4 Feature Groups

There are several sets of explanatory variables in the dataset:

Demographic Variables

- **SEX:** Gender (1 = Male, 2 = Female)
- **EDUCATION:** Education level (Graduate school, University, High school, Others, Unknown)
- **MARRIAGE:** Marital status (Married, Single, Others)
- **AGE:** Age in years

These variables act as contextual observations of the client's background that can be used to influence their financial behavior.

Billing Amount Variables

The variables correspond to the client's average credit card bill for six months.

- **BILL_AMT1** – September 2005
- **BILL_AMT2** – August 2005
- **BILL_AMT3** – July 2005
- **BILL_AMT4** – June 2005
- **BILL_AMT5** – May 2005
- **BILL_AMT6** – April 2005

All data are presented in New Taiwan Dollar (NTD). *These factors indicate the frequency and pace of consumption.*

Repayment Status Variables

These variables reflect previous repayment history and are important measures for determining financial stress and poor budget retention.

- **PAY_0** – Repayment status in September 2005
- **PAY_2** – Repayment status in August 2005
- **PAY_3** – Repayment status in July 2005
- **PAY_4** – Repayment status in June 2005
- **PAY_5** – Repayment status in May 2005
- **PAY_6** – Repayment status in April 2005

Repayment status values:

- -1 = Paid duly
- 1 = Payment delay for one month
- 2 = Payment delay for two months

…

- 8 = Payment delay for eight months
- 9 = Payment delay for nine months or more

A higher ratio means that the company pays off its payables more slowly and is less liquid in the short term.

Payment Amount Variables

These are the amounts that the client paid on their credit card balance.

- **PAY_AMT1** – Payment made in September 2005
- **PAY_AMT2** – Payment made in August 2005
- **PAY_AMT3** – Payment made in July 2005
- **PAY_AMT4** – Payment made in June 2005
- **PAY_AMT5** – Payment made in May 2005
- **PAY_AMT6** – Payment made in April 2005

Comparing payments made with the amount billed for indicates if clients tend to pay debts or under-pay their obligations.

2.3 Data Cleaning and Preprocessing

A number of preprocessing measures were taken in order to improve data quality without destroying meaningful financial behaviour:

- **Removal of non-informative features:**
  The ID was dropped as it is not useful for prediction.
- **Handling invalid categorical values:**
  Categories of schooling outside educational status (5 and 6) were classified in an "Other" category. Similarly, the invalid value for marriage status (0) was recoded as "Other". It reduced noise and eliminated misinterpretation.
- **Categorical Encoding:**
  Categorical features were integer-encoded in the uploaded data, and there was no need for further encoding.

- **Outlier Awareness:**
  Outliers in the billing, payment variables were included since they are realistic financial actions and important to demonstrates default risk.

These measures made the data compatible with supervised learning and preserved their behavioural representation.

2.4 Insights for Budgeting and Savings Optimisation

Exploratory analysis uncovered behaviour patterns that appeared to strongly correlate with the project's business goal. Those clients with many delays are substantially more likely to default, and high bills together with a low level of repayments was a poor sign for higher real lending than they could repay. These results inspired the development of engineered features to model repayment regularity, credit utilization and repayment history.

Originally intended for credit risk modelling, the data contains very strong behavioural indicators that are applicable to budgeting and optimising savings. Ethical concerns were considered, as predictions are meant to help users with transparent and explainable financial suggestions, not to form punitive judgements.

3. Feature Engineering Choices

Feature Engineering Feature engineering is important to convert raw financial data into relevant user behaviour indicators. The original dataset includes the precise financial transactions (billings, payments) but they should be summarised and reformulated to take account the budget control, debt sanity and potential savings.

The purpose of this phase is to design features, which describe:

- Consistency in debt repayment,
- The association of cost with repayment ability,
- Indicators of mounting financial pressure through the months.

These created features are not only meant to help the predictive power of the model, but also speed up feature importance ranking especially when used in conjunction with explainability methods like SHAP.

3.1 Aggregated Spending and Payment Features

For the purpose of summarizing six months of historical actions several aggregated and ratio features were constructed:

- **Average billing amount (avg_bill_amt)**
  Represents the user's average monthly consumption level and overall intensity of usage.
- **Average payment amount (avg_pay_amt)**
  Reflects the user's capacity to satisfy financial obligations that are recurrent.
- **Payment-to-bill ratio (payment_ratio)**
  Measures the air of what is billed and collected. Dark Blue is underpaying and poorly prioritising its debt.

Some features based on ratio contributed a small amount of invalid value in some cases where division by zero occurred. A cursing attempt of handling: median imputation and explicit reassignment (for the sake of backward-compatibility data-prep when updating to pandas versions >1) was used here.

In the aggregate, these characteristics form a brief profile of budgeting behaviour and repayment discipline.

3.2 Credit Utilisation and Risk Indicators

More financial pressure and credit-related risk features were engineered as follows:

- **Credit utilisation ratio (credit_utilization)**
  Determined as the average payment amount over the assigned credit limit. High utilization means one is over spending based on available credit capacity.
- **Maximum repayment delay (max_delay**
  The worst repayment status observed during the previous six months indicating severe liquidity constraints.
- **Average repayment delay (avg_delay)**
  Gives a more predictive, smoother and robust description of repayment behavior through time.

These indexes gauge both the depth and duration of financial stress.

3.3 Behavioural Trend Features

A feature based on a trend was introduced to recognize declining financial behaviour:

- **Debt trend (debt_trend)**
  It is measured by the most recent and oldest bill amounts. How do you measure this? If it is positive then you have a pattern of increasing debt over time, because clearly there's inadequate surplus if any at all.

This property allows the model to differentiate between stable and deteriorating financial paths.

3.4 Contribution to Budgeting and Savings Optimization

The engineered features enable us to move beyond raw transaction-level variables and the model can thus concentrate on patterns rather than single values. Customers demonstrating poor repayment ratios, the use of high credit limits and upward growth in debt levels can be actively identified at an early stage and provided with targeted advice on reduced discretionary spend, faster debt payoffs, increasing emergency savings.

Both in terms of predictive performance and practical relevance, this feature engineering approach enhances the model to be directly usable in a budgeting/savings optimisation context.

4. Model Development

The modeling task is to predict whether a user is highly likely to suffer financial tension, i.e. repay the credit card bill disfavored in the next month. On the business side of things, this means that the fintech could make advance suggestions about how people should adjust their spending (ie allocating larger proportion of one's salary toward debt repayment and savings).

This task is treated as a binary classification with the target variable to represent default (1) or non-default (0).

4.1 Train–Test Split

The dataset has been divided into training and testing data using 80/20 splitting. The stratification was used to maintain the proportion between different classes of target, which is crucial as non-default and default cases are imbalanced.

It also guarantees a fair evaluation of model performance in a representative and unbiased test set.

4.2 Baseline Model: Logistic Regression

Logistic Regression was chosen as the starting model because it is simple, interpretable and one of the most commonly employed algorithms in credit risk estimation. It acts as the litter bin and benchmark for complex models.

The reason for the selection of this model is that it:

- Is popular in financial risk applications,
- Generates probabilistic predictions that are relevant for risk informed decision making,
- Offers easily interpretable coefficients,
- The one-move analysis can serve as a strong and significant benchmark.

4.3 Primary Model: Random Forest Classifier

The model we have chosen to use is the Random Forest Classifier, due on its basis to capturing complex financial behaviours.

Random Forest was selected because it,

- When capturing non-linear relationships between financial variables,
- Effectively models interactions between features,
- Is resistant to noise and outliers,
- Performs well on tabular financial data,

- Orders The order in which any SHAP grabber(s) is executed.

These properties make Random Forest well-suited for modelling budgeting and repayment behaviours that cannot be effectively described by a linear model.

4.4 Feature Selection and Scaling

All relevant numerical and encoded categorical variables were included in model training. Feature scaling was applied where required for Logistic Regression to ensure numerical stability and optimal performance.

In contrast, Random Forest does not require feature scaling, allowing it to operate directly on the engineered features without additional transformation.

4.5 Model Training Overview

Both models were trained using the same training dataset to ensure a fair and consistent comparison. Their performance is evaluated using metrics appropriate for imbalanced financial classification problems, including recall, F1-score, and ROC-AUC, which better reflect the business objective of identifying financially at-risk users.

5. Results and Evaluation

Because having ill-intentioned users get access to your serviceman cause a significant financial loss, evaluation of model focuses more on the chances to be rightly capable of identifying high-risk users, and not only accuracy. In a budgeting and savings optimization scenario not identifying someone "at risk" of default is missing the chance to intervene.

Therefore the subsequent measurements are favoured:

- **Recall (Default = 1):** how well the model can identify users who are under financial stress,
- **F1 score:** trade off between precision and recall,
- **ROC-AUC:** global discrimination power of the model,
- **Confusion Matrix:** summary of classification results.

Random Forest had a lower overall accuracy but 3.2x higher recall for defaults (default rate probability) than Logistic Regression (where default cases yield very low recall levels of about 0.24 and, with our setup, less meaningful models that perform better in terms of accuracy). This makes Random Forest more suited to pinpoint users who are at financial risk.

5.1 Confusion Matrix Analysis

The confusion matrix helps to understand various types of classification results:

- **True Positives:** users that were correctly classified as risky,
- **False Negatives:** misclassified high-risk users as low risk,
- **False Positives:** users marked as high-risk but who do not actually default.

In terms of business impact, the failure to correctly predict users at risk is the error that matters most (of course).

5.2 ROC Curve and AUC

The ROC curve is a graph that shows the trade-off between true positive rate and false positive rate at every classification threshold. The higher the Area Under the Curve (AUC) the better is to differentiate between default and non-default users.

For this problem ROC-AUC is particularly appropriate because of the class imbalance nature of dataset and a strong, threshold- agnostic measure of overall model performance.

5.3 Model Comparison

Both models were tested on the same testing data to make a fair comparison. In addition to that, Logistic Regression provides a transparent and interpretable baseline model but lacks the ability to learn complex behavioural patterns of financial stress.

Contrary, the Random Forest model have higher recall and ROC-AUC, that means he is better identifying users with default risk. This comparative advantage justifies its use as a leading model for proactive budgeting and savings optimisation.

5.4 Business Interpretation of Results

The results of the assessments show the potential for machine learning to assist in early detection of individuals suffering from financial strain. With the Random Forest model in place, the fintech platform can act proactively to:

- Identify flag users who may benefit from discretionary spending reduction,
- Recommendation: Debt Repayment This costs more as compared to money put in the bank, but this step of yours is fully funded by average investors.
- Foster the development of emergency savings cushions.

These findings are a way to convert predictive outputs into financial action-points, which drive responsible decisions and increase user trust in automated advice.

6. SHAP Explainability

Performance on its own doesn't cut it in the world of finance. Users, regulators and financial institutions need to be able to understand why a model gives a particular prediction. That's why AI must be Explainable: to build trust, fairness and provide insight into the decision.

We interpret the predictions of the base Random Forest model in this project using SHAP (SHapley Additive exPlanations). SHAP explanation can be used to explain in depth one customer but also the average behaviour across customers.

Why SHAP Was Selected

SHAP is chosen with the following advantagesconvenient-Because while eligible:

- It is based on cooperative game theory, so to guarantee the rationality of the feature attribution.
- It gives you stable and additive explanations over predictions,
- It is fit for complicated non-linear models such as Random Forest,
- It creates visualisations that are easy to understand for non-technical people.

These characteristics also render SHAP a favorable choice for fintech applications where both prediction accuracy and interpretability are needed.

6.1 Global Explainability

Global SHAP reveals which features contribute most to default risk for the population of users. The findings reveal repayment and credit utilisation as major determinants of financial stress akin to conventional risk indicators.

This consistency of SHAP feature importance with prior financial behavior patterns gives us confidence that the model is picking up on real signals, not spurious correlations. At the business end, there are global explanations as to why decisions about products - what could be common risk drivers, and targeted financial interventions.

6.2 Local Explainability

Local SHAP explanations are for single predictions, describing how each feature contributes to the risk score of a particular user. This level of detail is useful for personalised budgeting advice because it makes it possible to explain recommendations in a clear and intuitive way.

For instance, a customer might be high risk due to some combination of high credit utilization, low payment-to-bill ratios and recent late payments. SHAP supports explaining the contributing factors explicitly, and enables users to comprehend why the recommended interventions are reasonable.

6.3 Business Value of Explainability

By incorporating SHAP explanations into the modelling process, the fintech platform can:

- Reinforce confidence of users in automated recommendations,
- Advocate for responsible and transparent financial advice,
- Provide actionable insights and not blackbox predictions,
- Aid compliance with ethical and regulatory requirements regarding explainable decision making.

All in all, explainability turns the model from a black box predictor into a decision support tool that links technical outputs to user-desired financial results.

7. Reflection and Limitations

This work presented an applied example of machine learning methods to a real-world fin-tech problem, budgeting and savings optimisation. Two types of classification models were built and compared including Logistic Regression (as base model) and Random Forest (as primary model).

Logistic Regression performed with slightly better accuracy (around 0.81 overall), but was poorer in identifying financially stressed users. Low recall in default case resulted in high number of high risk users being classified as low-risk. From a business standpoint, this is a crucial constraint as these users would never get timely budgeting or savings interventions.

On the other hand, the Random Forest model found a better trade off between performance indicators and metrics that tracks business goals. Despite slight worst overall accuracy, it showed much higher recall and f1-score for the default class and around a better ROC-AUC (about 0.78). This validates its stronger capacity to differentiate users that are in danger of financial stress, which is more important in proactively guiding people financially than mere increased accuracy. These findings emphasize the need to consider suitable metric if it corresponds more accurately to business values instead of accuracy.

7.1 Insights from Explainability Analysis

Explainability based on SHAP had a pivotal role in validating the model's predictions and associating them with actual financial behavior. A Global SHAP analysis showed that variables related to repayment, for example the maximum delay in payment and recent status of repayment were the most important factors determining likelihood of default. Other features, like poor budget allocation, also added substantial value through their co-display alongside financial stress.

SHAP local explanations also demonstrated how combinations of delayed payment, high credit utilisation, and missed repayments influenced individual predictions for default. These results are consistent with financial intuition, and raise confidence that the model captures relevant behavioural patterns instead of merely spurious correlations. On a fintech level, it allows the business to provide responsible financial guidance and builds trust with the user.

7.2 Limitations of this Work

There are some limitations to this research as well, however.

- **Class Imbalance:**
  There is a severe class imbalance in the dataset, which complicates both training and evaluation of the models. Although the class weighting was used, however some bias toward a majority class persists.

- **Proxy Target Variable:**
  Credit card default acted as a proxy for bad budgeting and saving behaviour. Although this is a plausible estimation, default does not present the full spectrum of financial well-being, including informal savings, income volatility and external financial support.
- **Static Snapshot:**
  Data presents historical activity throughout a specific period in time. In practice, financial behaviour is evolving over time and can be better characterised by time series compared with static models.
- **Threshold Dependency:**
  Predictions made by the model rely on a threshold that is set. Tuning this threshold further would help fine tune the balance between false positives and false negatives for business purposes and risk acceptance.

7.3 Future Improvements

There are a few extensions that could improve the robustness and generality of this work:

- Including more financial features like income information or spending categories at a transaction level,
- Utilize state of the art solutions to treat class imbalance such as, SMOTE and cost-sensitive learning algorithms.
- We could look into dynamic or sequential models to better accommodate behaviour trends,
- Embedding the model in a real time budgeting platform with continuous feedback loops,
- Broader Deploying explainability on user facing dashboards with simplified SHAP-based explanations.

Conclusion

All in all, this project establishes a successful model to prove that machine learning techniques can be used to aid in budgeting and optimizing for savings when paired with careful feature engineering, the right evaluation metrics, and methods for interpretability. The Random Forest model, with SHAP-based explanations, offers a solid and interpretable approach for profiling at-risk users to empower proactive financial guidance.

On a more general level, this paper demonstrates the need to trade off predictive performance for interpretability in financial domain, where transparency and user trust are as important as accuracy. The project showcases how technical modelling decisions can be combined with business and ethical concerns, demonstrating the potential of machine learning as a useful decision support tool for fintech businesses.

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